

Cellular Biophysics (BIO516) – Quiz 1

Marks: 20, Time: 1 hr

Date: 23/03/2026

Instructions to candidates

1. This paper consists of three sections: Section A (Multiple Choice: Q1-Q10), Section B (Fill in the Blanks: Q11-Q15), and Section C (Numerical Problems: Q16-Q18).

2. **Answer all the questions**

3. Use of calculator is permitted.

4. Standard Constants:

- Gas Constant (R) = $8.315 \text{ J}/(\text{mol} \cdot \text{K})$
- Faraday's Constant (F) = $96,485 \text{ C/mol}$
- Temperature (T) = 298 K (unless specified)

Section A

[Marks: 10 X 1]

Q1 In P-type ATPases, what does the "P" stand for?

- A) Potassium transport.
- B) Passive diffusion.
- C) Reversible phosphorylation of the transporter during the catalytic cycle.**
- A) Proton pumping only.

Q2. Which of the following best describes the "disequilibrium" that allows ATP to serve as the major carrier of chemical energy?

- A) Cells maintain ATP at its exact equilibrium concentration.
- B) Cells maintain a concentration of ATP far above its equilibrium concentration.**
- C) ATP is thermodynamically stable but kinetically unstable.
- D) The concentration of ATP is kept lower than ADP to drive reactions.

Q3. A transport process is termed "electrogenic" when:

- A) It moves a solute down its concentration gradient.
- B) It moves an ion without an accompanying counterion, producing an electrical potential.**
- C) It uses a symporter to move two neutral molecules.
- D) It involves the hydrolysis of GTP instead of ATP.

Q4. Which factor contributes to the large negative ΔG of ATP hydrolysis by relieving electrostatic repulsion?

- A) The formation of resonance forms of P_i .
- B) The ionization of ADP^{2-} to ADP^{3-} .
- C) The separation of one of the three negatively charged phosphate groups.**
- D) The binding of Mg^{2+} to the adenine base.

Q5 SERCA is a classic example of which transport mechanism?

- A) **Primary active transport**
- B) Secondary active transport
- C) Facilitated diffusion
- D) Scalar activation by Glucose

Q6. Subject to pumps, "Static Head" is defined as:

- A) The flow rate when the electrochemical potential difference is zero.
- B) **The opposing electrochemical potential difference that reduces net flux to zero.**
- C) The maximum velocity of the pump.
- D) The structural weight of the protein.

Q7 If metabolic inhibitors are added to a cell, why does secondary active transport often continue briefly while primary active transport stops immediately?

- A) Secondary transporters possess internal ATP reservoirs.
- B) **Secondary transport relies on a pre-existing electrochemical gradient of a co-solute that takes time to dissipate.**
- C) Secondary transporters are structurally more stable than ATPases.
- D) Primary transport is actually a passive process.

Q8 Why is the $\text{Mg} \cdot \text{ATP}^{2-}$ complex, rather than free ATP^{4-} , the functional substrate for most biophysical reactions in the cell?

- A) Magnesium increases the total negative charge of the molecule.
- B) **Due to high cellular Mg^{2+} concentrations, ATP exists largely in this complex form to participate in phosphate transfers.**
- C) Free ATP is too thermodynamically stable to react.
- D) Magnesium acts as the primary chemical energy source.

Q9 In a scalar transport model, how does metabolic coupling drive the transport cycle?

- A) It physically pushes the carrier through the lipid bilayer.
- B) **It shifts the equilibrium of the activation reaction, raising the ratio of "transport-ready" carriers on one side.**
- C) It utilizes thermal gradients to move solutes.
- D) It permanently deactivates the carrier after a single transit.

Q10 In secondary active cotransport (symport), what occurs if the substrate (A) is removed from the cis side?

- A) The flux of the co-solute (B) increases significantly.
- B) **The flux of the co-solute (B) decreases because the carrier requires both to transit effectively.**
- C) The co-solute immediately begins moving in the opposite direction.
- D) The ATP hydrolysis rate at the primary pump increases.

Q11. The P-type ATPase mechanism requires the protein to alternate between two main conformations: the inward-facing _____ and the outward-facing _____. (**Ans: E1 and E2**)

Q12. In P-type ATPases, the _____ domain is not responsible for the conformational changes that occur during the catalytic cycle. (**Ans: P or N**)

Q13 In secondary active transport, the solute whose generated gradient drives the flux of another species is called the _____. (**Ans: Cosolute**)

Q14. The _____ pump in animal tissues maintains the cytosolic Ca^{2+} level below $1\ \mu\text{M}$. (**Ans: plasma membrane calcium (or SERCA)**)

Q15. The electrochemical potential difference ($\Delta\mu_i$) is known as the _____ driving force for the passive flux of a species. (**Ans: conjugate**)

Section C

Q16. Using the stoichiometric relationship $J_B = \nu_B J_A$, find the flux of cosolute B if the coupling coefficient is 2 and the substrate flux is $5\ \text{mmol/s}$. [Marks: 1]

Answer: $J_B = 2 \times 5 = 10\ \text{mmol/s}$.

Q17. A researcher is studying the thermodynamic feasibility of a primary active transporter in an E. coli cell at $37\ ^\circ\text{C}$. The transporter moves a specific cation "X" from the cytosol to the extracellular fluid with a stoichiometry of 2 ions per ATP hydrolyzed.

Given: Intracellular Conditions (E. coli): $[\text{ATP}] = 7.90\text{mM}$, $[\text{ADP}] = 1.04\text{mM}$, and $[\text{Pi}] = 7.9\text{mM}$. Standard Free Energy: ΔG° for ATP hydrolysis is -30.5kJ/mol . Transport Work: The calculated energy required to move the ions against the existing electrochemical gradient is $\Delta G_{\text{transport}} = +48.2\text{kJ/mol}$.

Calculate the ΔG_{total} , and state whether the transport of 2 ions of "X" per ATP is feasible or not feasible. [Marks: 2]

Answer Key

1. Actual Free Energy ΔG_p :

$$\Delta G_p = -30.5\ \text{kJ/mol} + (0.008315 \times 310) \ln \frac{(1.04 \times 10^{-3})(7.9 \times 10^{-3})}{(7.9 \times 10^{-3})}$$

$$\Delta G_p = -30.5 + (2.58)(-6.87)$$

$$\Delta G_p \sim -48.22\ \text{kJ/mol}$$

2. Net Free Energy (ΔG_{total}): $\Delta G_p + (2\Delta G_{\text{transport}}) = (-48.22 + 96.4)\ \text{kJ/mol} = +48.18\ \text{kJ/mol}$

- **Feasibility:** Not feasible as the $\Delta G_{\text{total}} > 0$.

Q18. A specialized epithelial cell at 37 °C utilizes a Na⁺-glucose symporter to accumulate glucose from the extracellular fluid. In this specific tissue, the pump operates with a coupling stoichiometry of **2 Na⁺ ions** transported for every **1 glucose molecule**.

Given Data:

- **Extracellular (Phase I):** Na⁺_{out} = 145 mM; Glucose_{out} = 5mM.
- **Intracellular (Phase II):** Na⁺_{in} = 12 mM.
- **Membrane Potential:** $\Delta\phi = 70$ mV (interior negative).
- **Constants:** R = 8.315 J/(mol·K); F = 96,485 C/mol.

Calculate the maximum theoretical intracellular glucose concentration [glucose]_{in} that can be achieved, assuming the pump reaches the thermodynamic limit. [Marks: 2]

Answer:

1. **Driving Force ($\Delta\tilde{\mu}_{Na}$):**

- Chemical component: $RT \ln(145/12) = (8.315 \times 310) \times \ln(12.08) \approx 6,432$ J/mol.
- Electrical component: $F\Delta\psi = 96,485 \times 0.070 \text{ V} \approx 6,754$ J/mol.
- **Total $\Delta\tilde{\mu}_{Na} \approx 13,186$ J/mol (or 13.19 kJ/mol).**

2. **Maximum Glucose Accumulation:**

- Using stoichiometry $v_{Na} = 2$: $RT \ln([G]_{in}/[G]_{out}) = 2 \times 13,186 = 26,372$ J/mol.
- $\ln([G]_{in}/5) = 26,372/(8.315 \times 310) \approx 10.23$.
- $[G]_{in}/5 = e^{10.23} \approx 27,722$.
- **Final $[Glucose]_{in} \approx 138,610$ mM.**